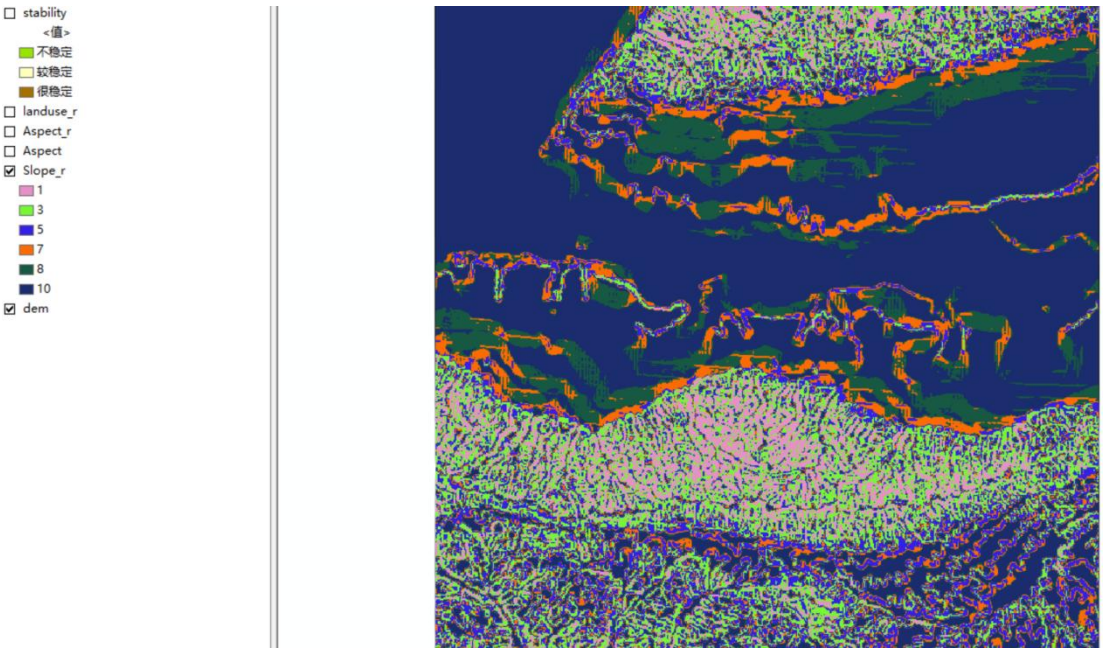
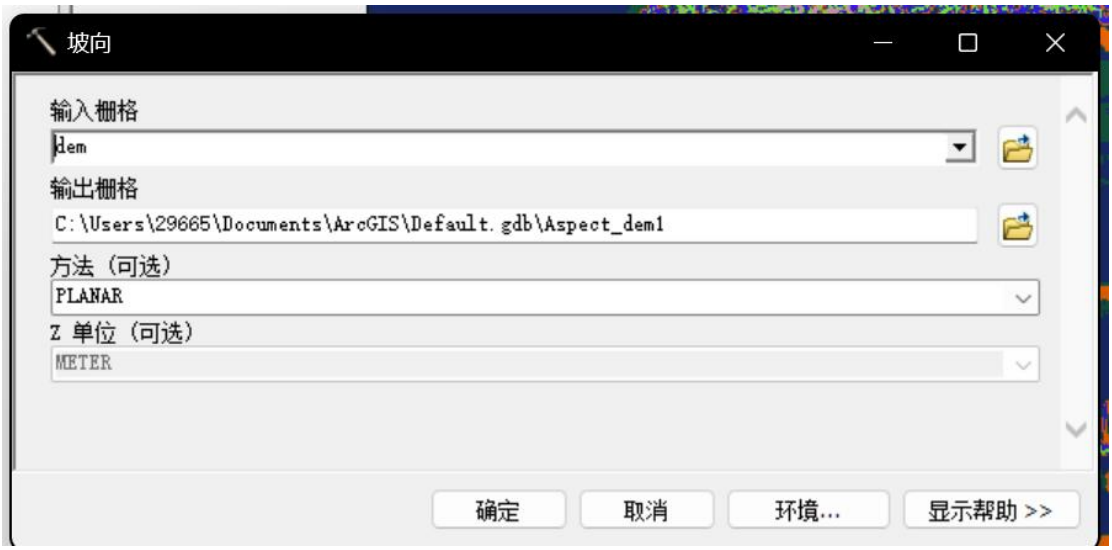
# 练习一

（1）提取坡度数据。

（2）坡度数据重分类

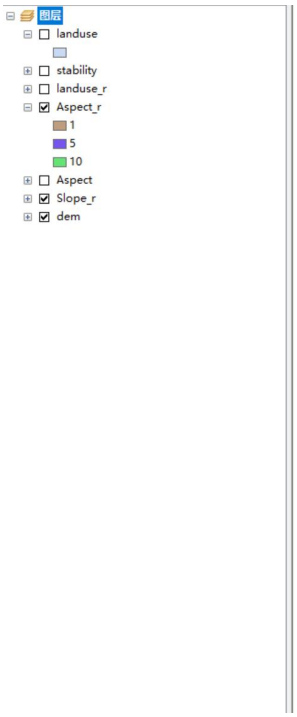

stability
<值>
不稳定
较稳定
很稳定
landuse_r
Aspect_r
Aspect
Slope_r
1
3
5
7
8
10
dem

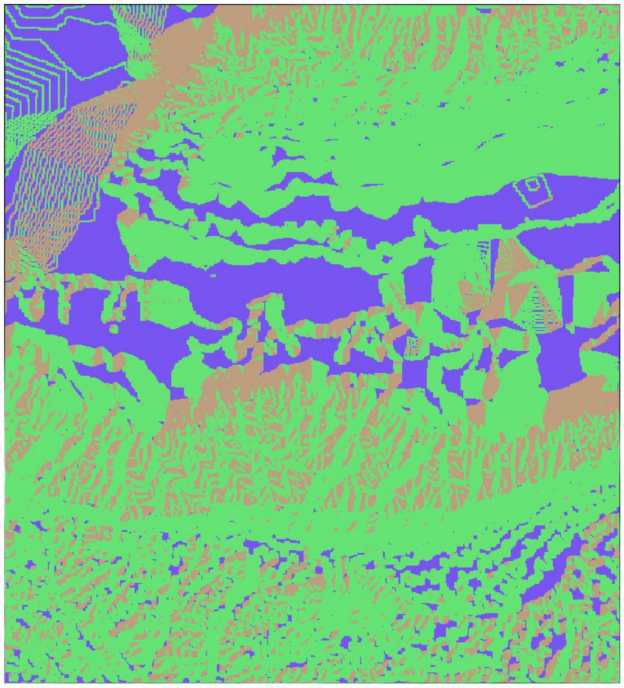
（3）提取坡向数据


坡向
输入栅格
dem
输出栅格
C:\Users\29665\Documents\ArcGIS\Default.gdb\Aspect_dem1
方法（可选）
PLANAR
Z 单位（可选）
METER
确定
取消
环境...
显示帮助 >>

(4)坡向数据重分类

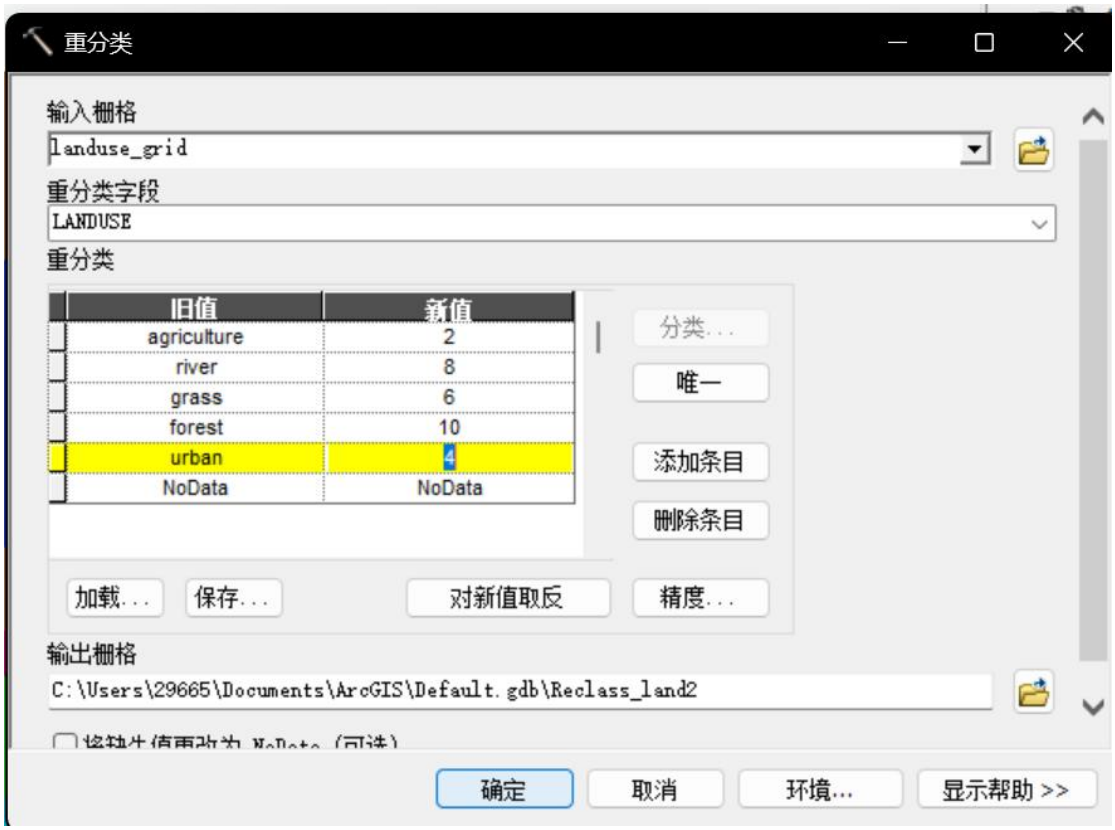
重分类
输入栅格
Aspect
重分类字段
Value
重分类
旧值
新值
-1
5
0 - 90
10
90 - 270
10
270 - 360
1
NoData
NoData
分类...
唯一
添加条目
删除条目
加载...
保存...
对新值取反
精度...
输出栅格
D:\GeoDataBase\GIS_Application\空间分析综合练习\新建文件地理数据库.gdb\Aspect_r
将缺失值更改为 NoData (可选)
确定
取消
环境...
显示帮助 >>

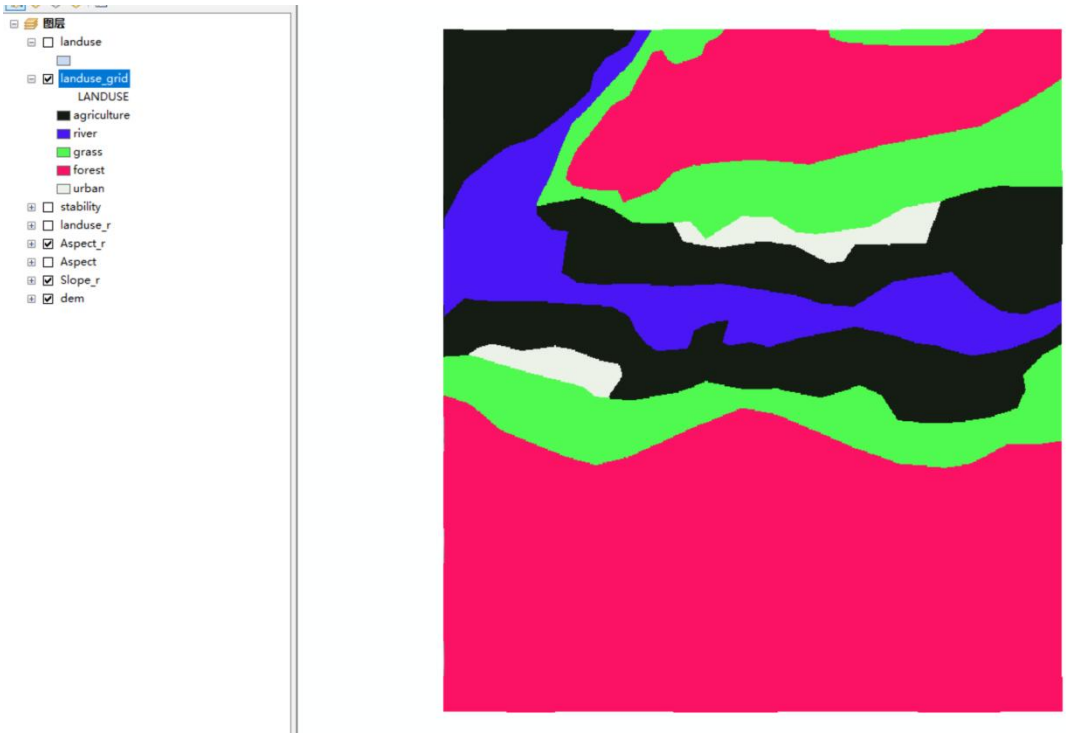

图层
landuse
stability
landuse_r
Aspect_r
1
5
10
Aspect
Slope_r
dem



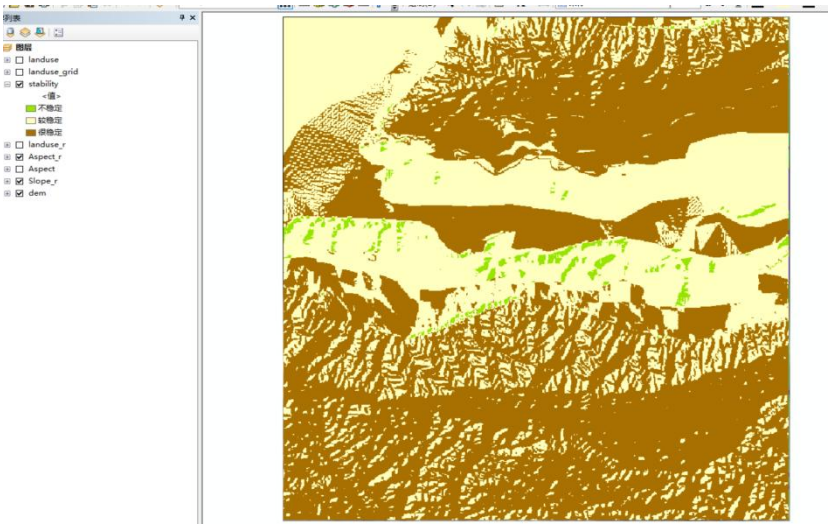
(5)将 landuse 依据土地利用类型转换为栅格数据

（6）土地利用重分类

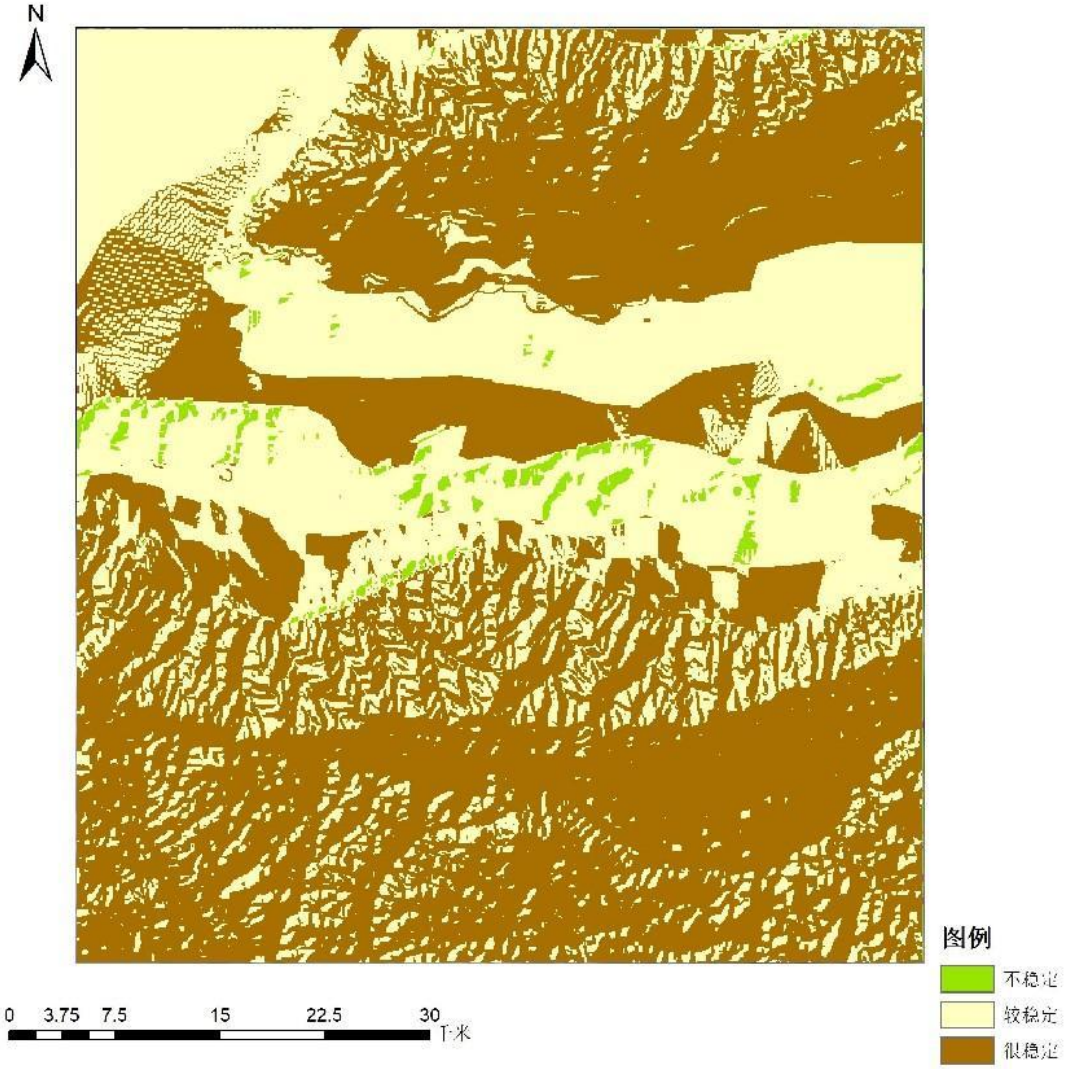


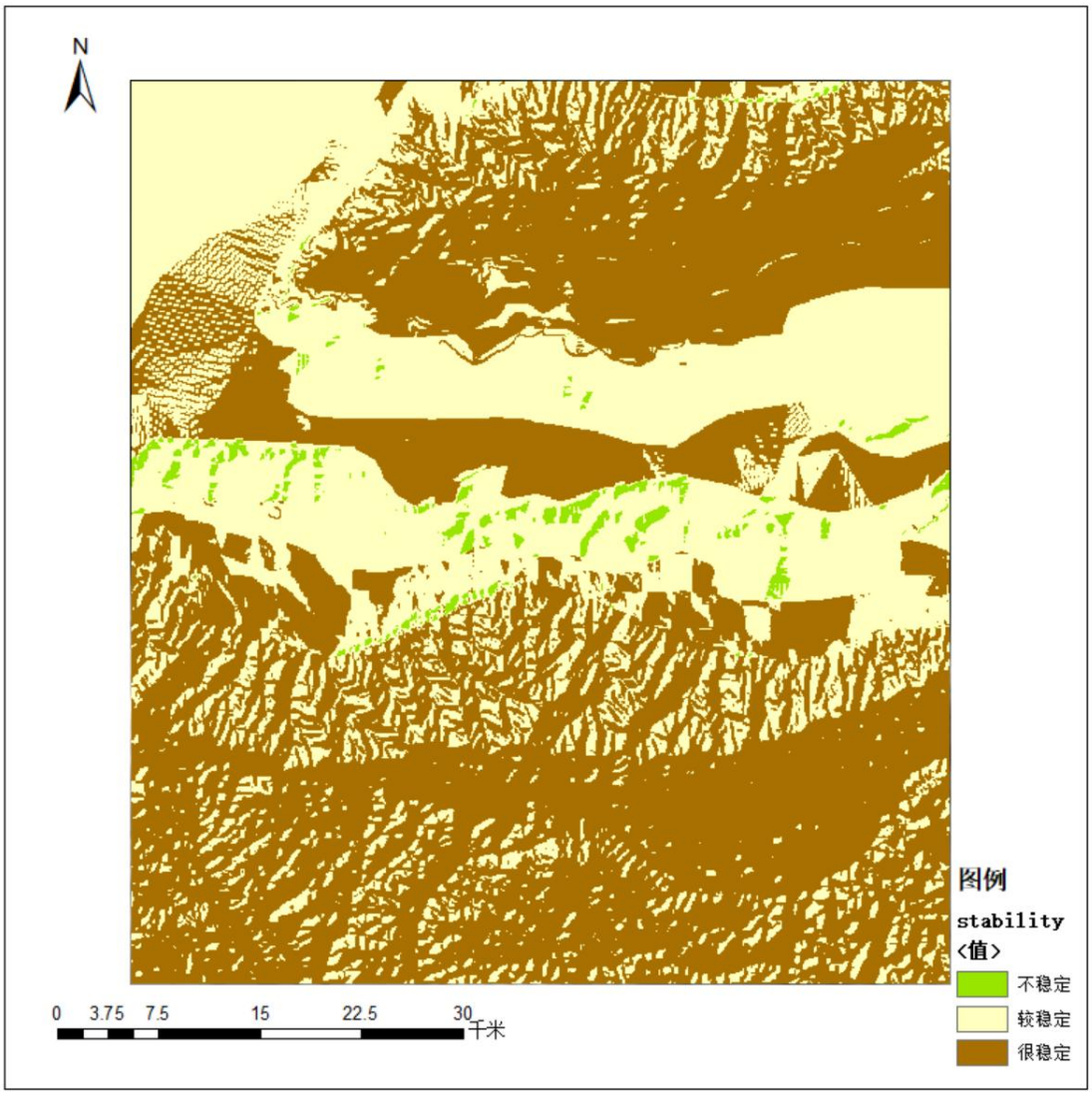



（7）综合考虑坡度、坡向和土地利用类型进行空间叠加分析



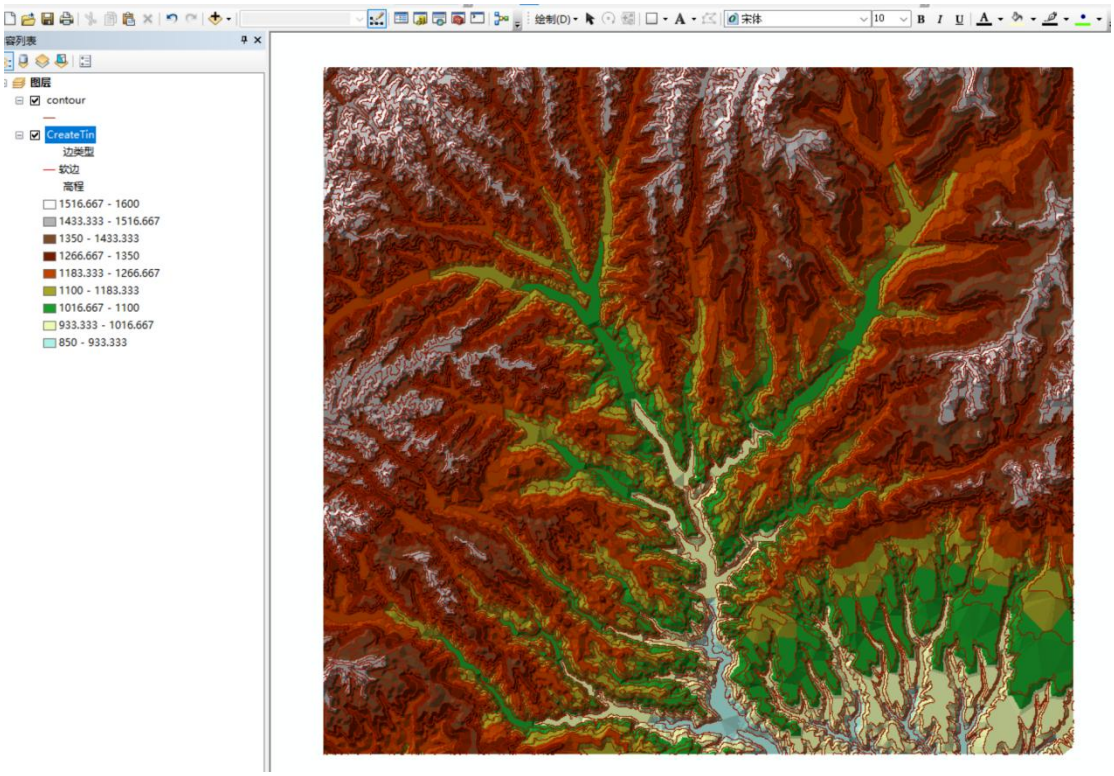
(8)将叠加结果按照土壤稳定性等级进行显示




N
0 3.75 7.5 15 22.5 30 千米
图例
stability
<值>
不稳定
较稳定
很稳定

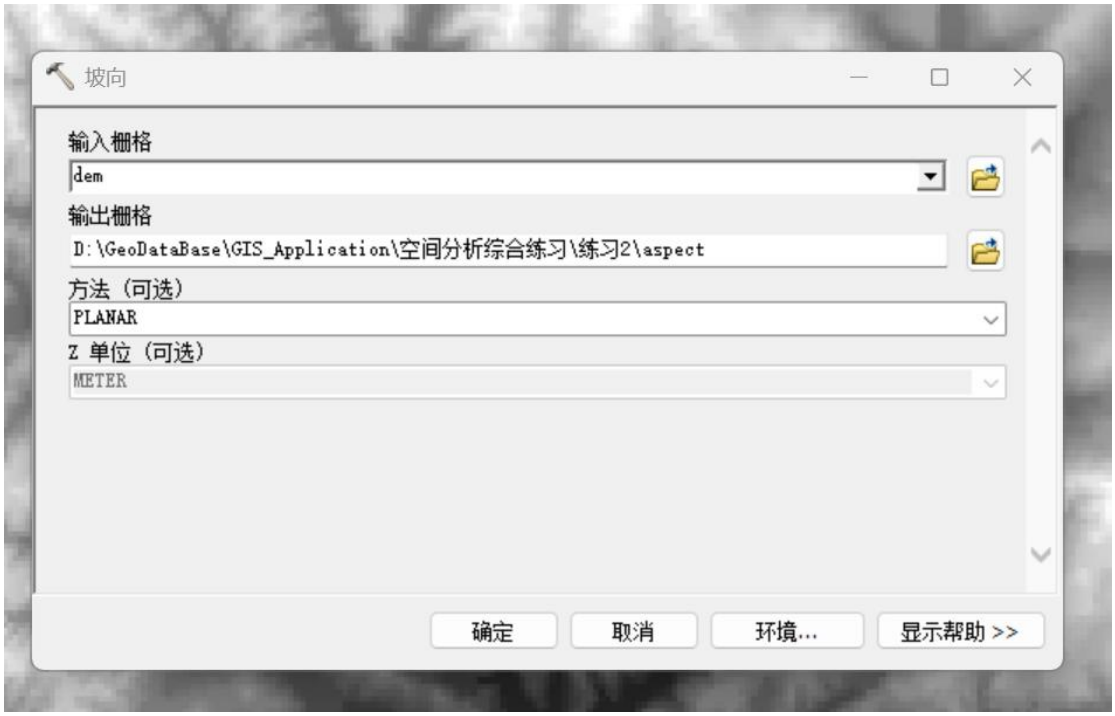
练习二

（1）利用等高线数据，构建 DEM。

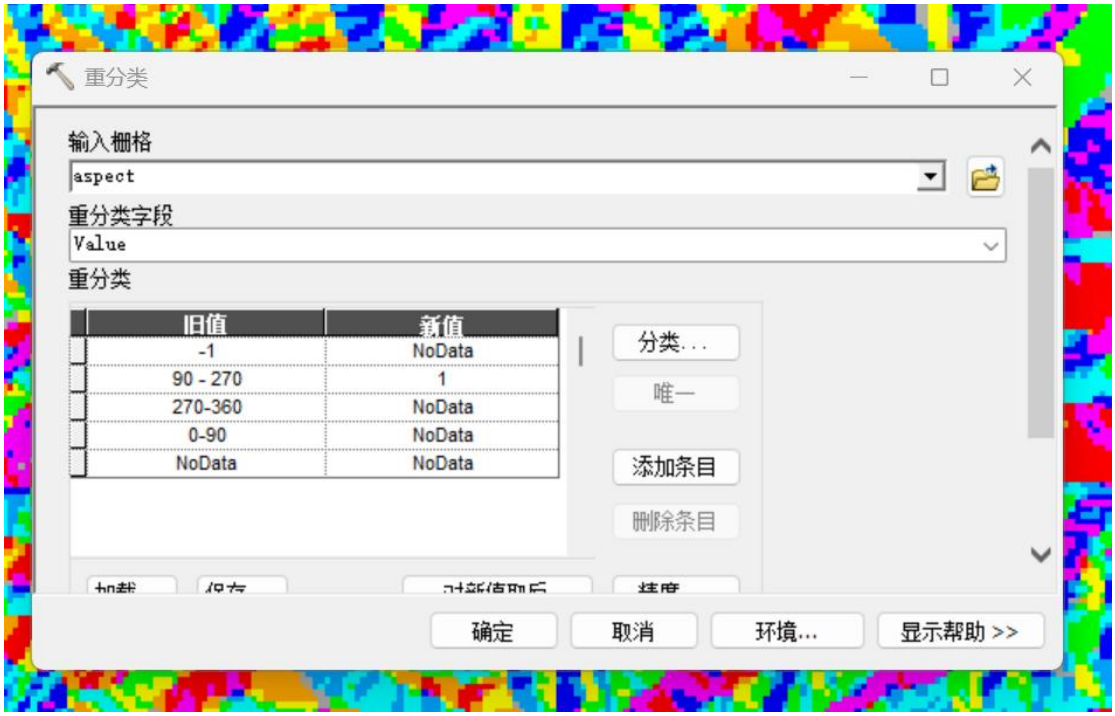


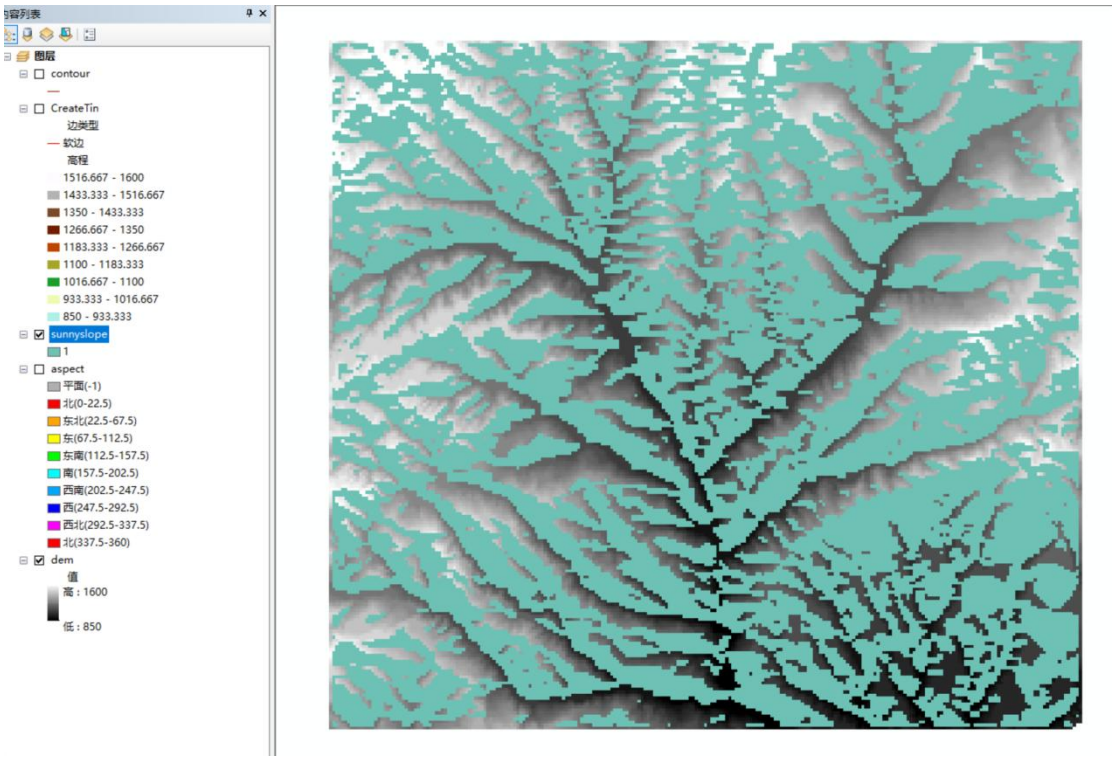

⑵ 将 TIN 转换成格网 DEM，以便于进行表面分析和与其他数据的叠加分析。

(3) 提取坡向数据



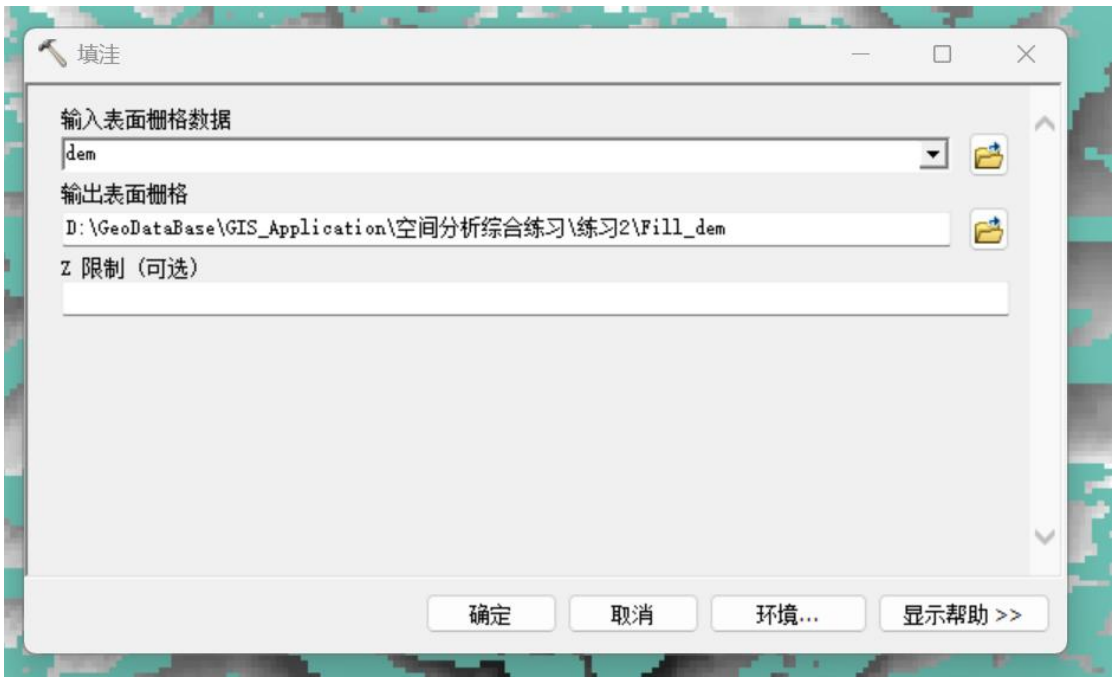

(4) 提取阳坡数据。





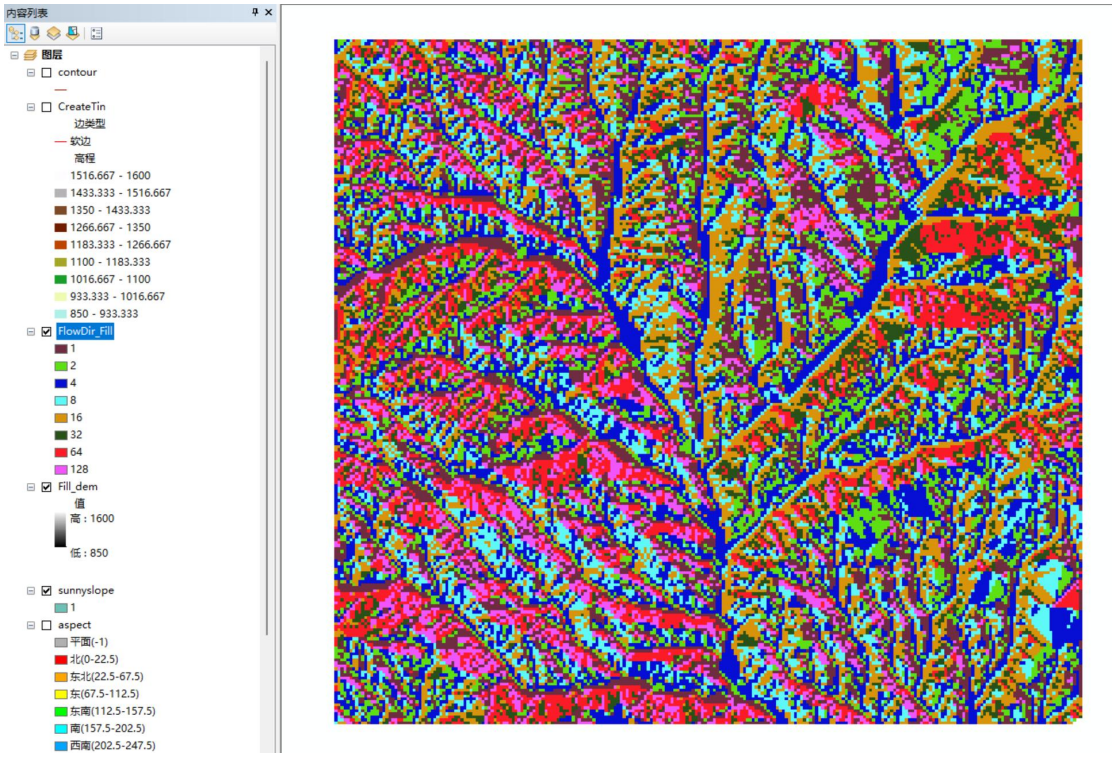

## (5) 对 DEM 数据进行洼地填充处理。

① 选择【spaitial analyst tools】|【Hydrology】|【Fill】工具。

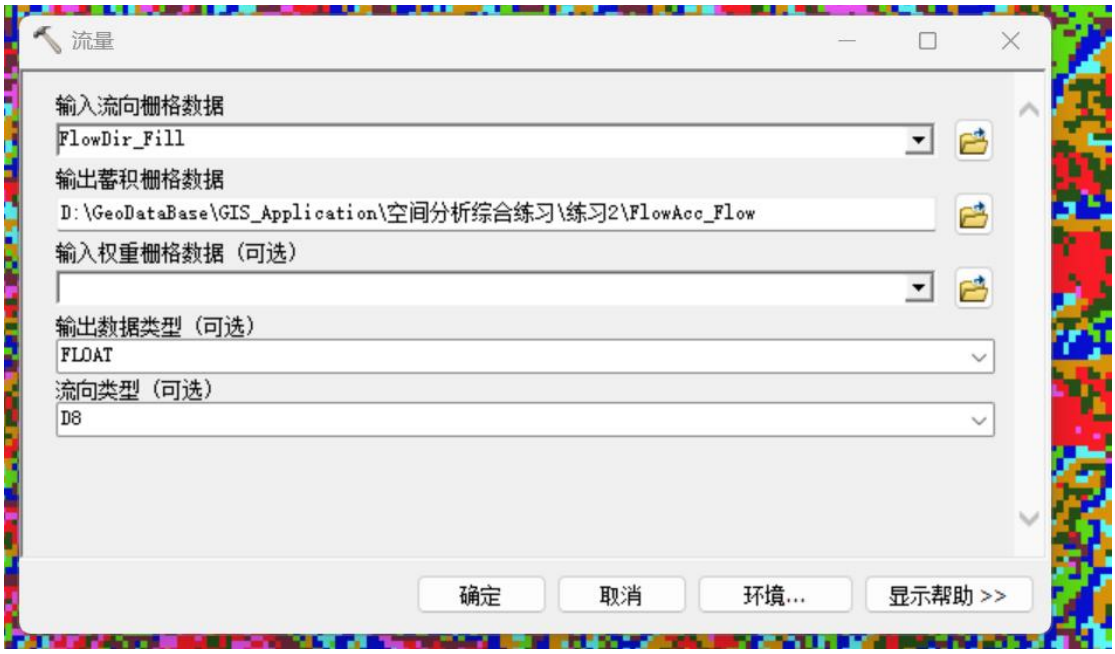


② 提取水流流向数据

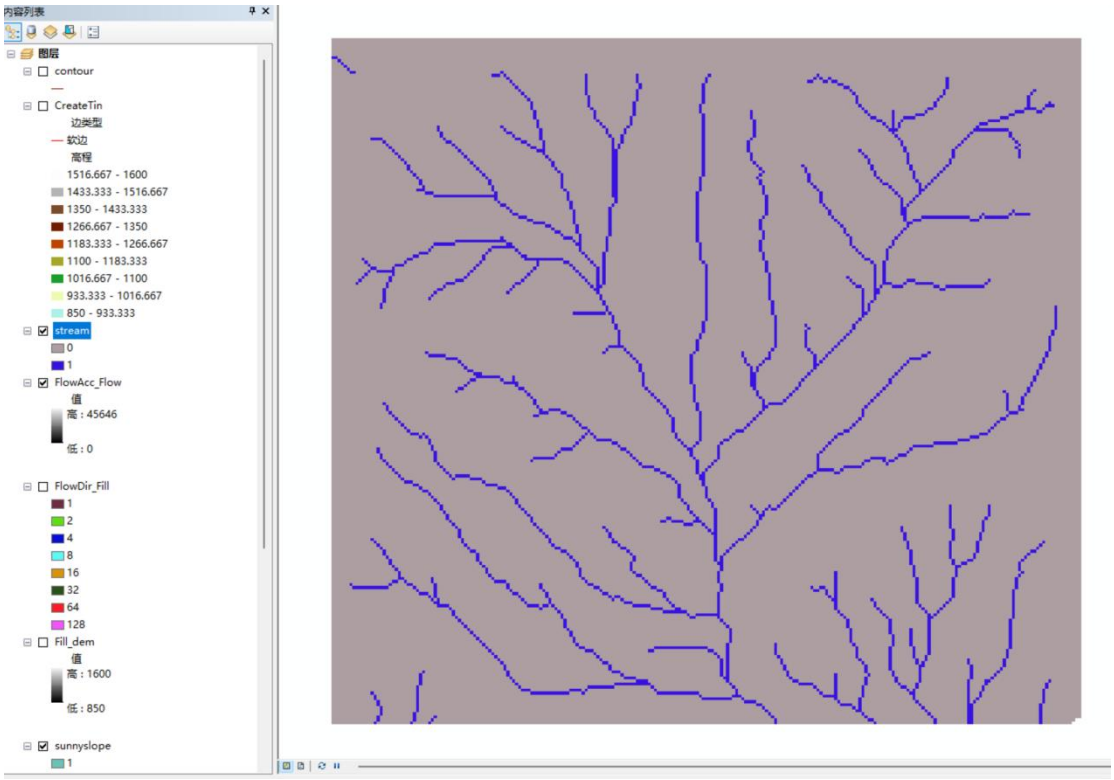
流向
输入表面栅格数据
Fill_dem
输出流向栅格数据
D:\GeoDataBase\GIS_Application\空间分析综合练习\练习2\FlowDir_Fill
强制所有边缘像元向外流动（可选）
输出下降率栅格数据（可选）
流向类型（可选）
D8
确定
取消
环境...
显示帮助 >>


内容列表
图层
contour
CreateTin
边类型
软边
高程
1516.667 - 1600
1433.333 - 1516.667
1350 - 1433.333
1266.667 - 1350
1183.333 - 1266.667
1100 - 1183.333
1016.667 - 1100
933.333 - 1016.667
850 - 933.333
FlowDir_Fill
1
2
4
8
16
32
64
128
Fill_dem
值
高 : 1600
低 : 850
sunnyslope
1
aspect
平面(-1)
北(0-22.5)
东北(22.5-67.5)
东(67.5-112.5)
东南(112.5-157.5)
南(157.5-202.5)
西南(202.5-247.5)

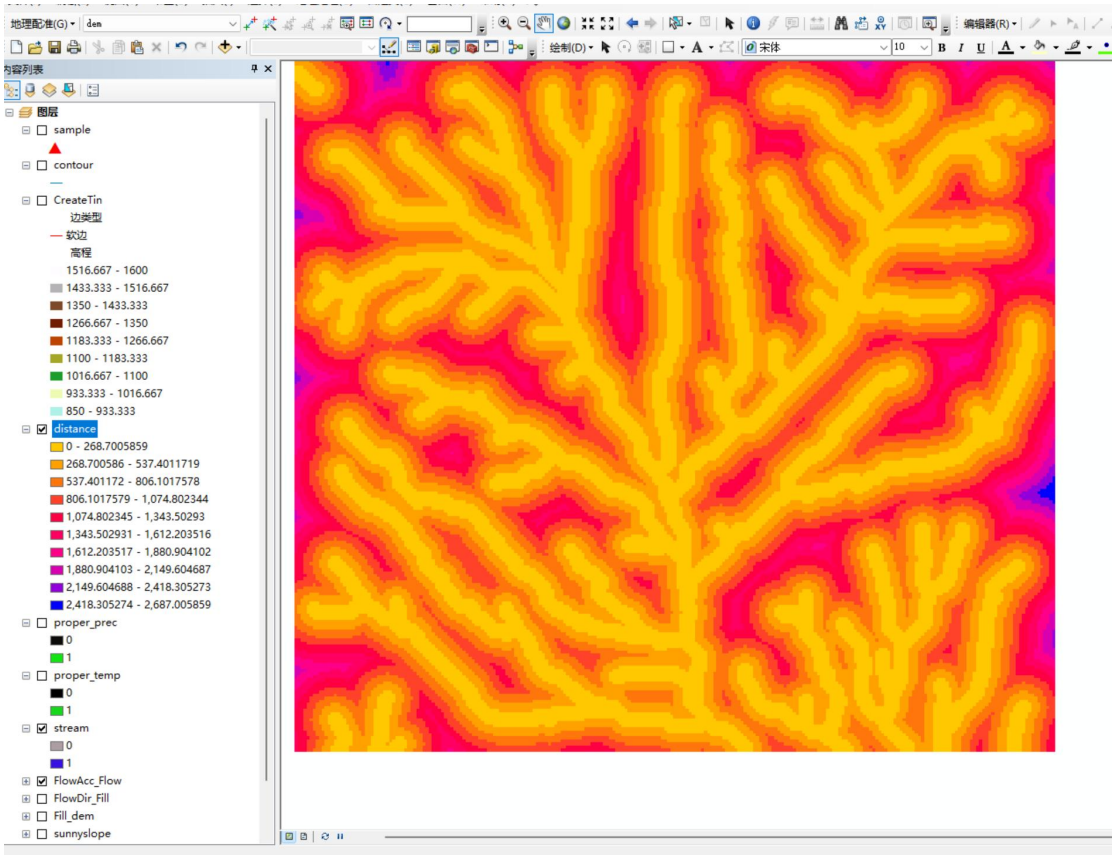
③ 提取汇流累积量数据

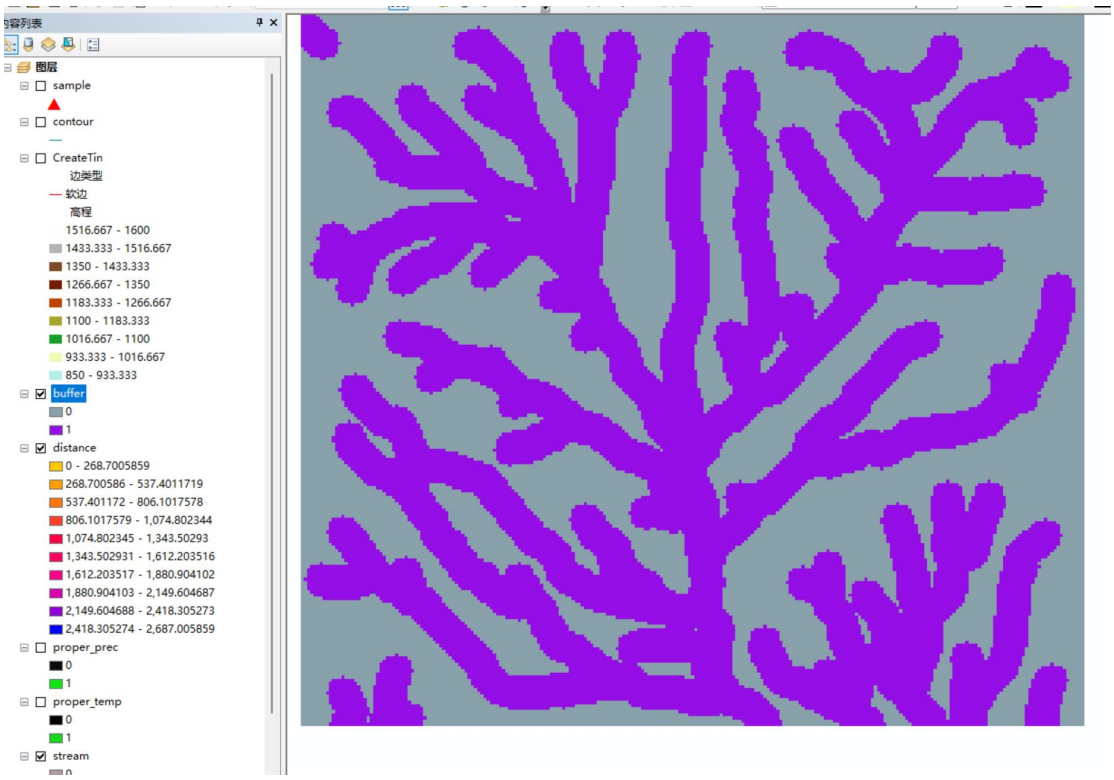



④ 提取沟谷网络

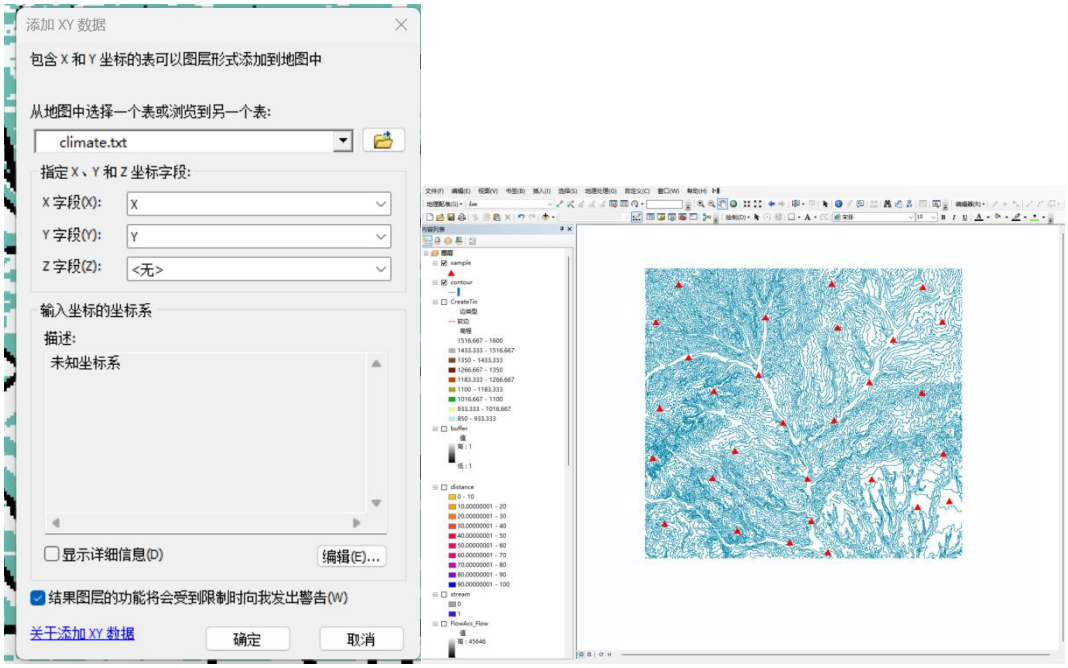


## (6) 用水系数据进行缓冲区分析

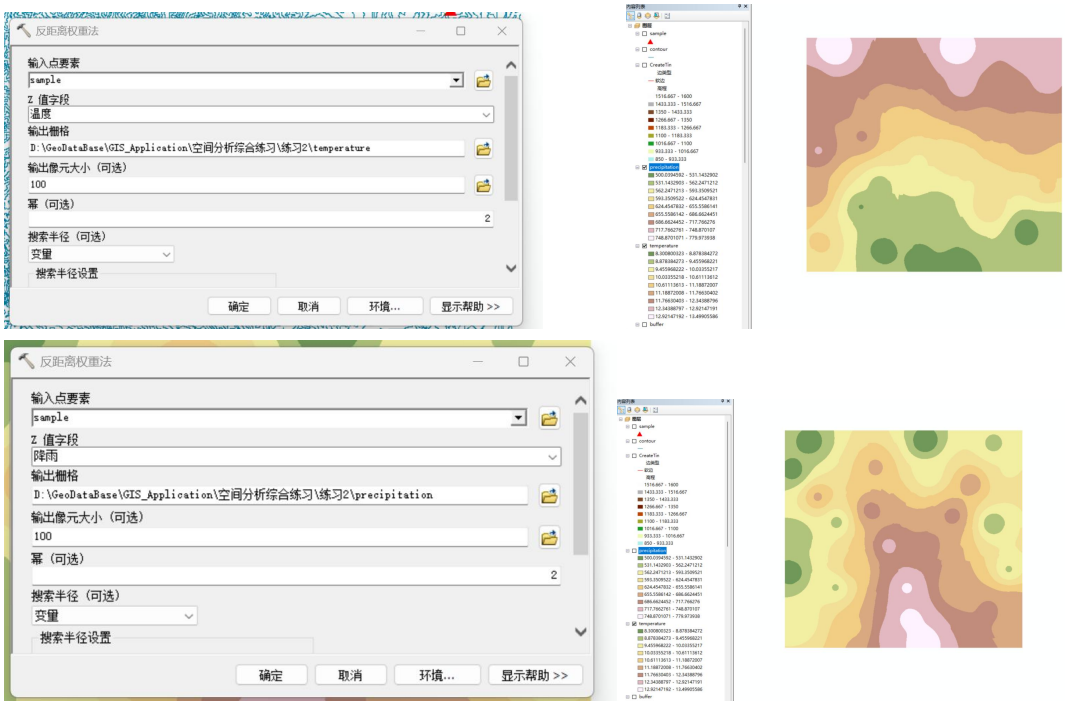


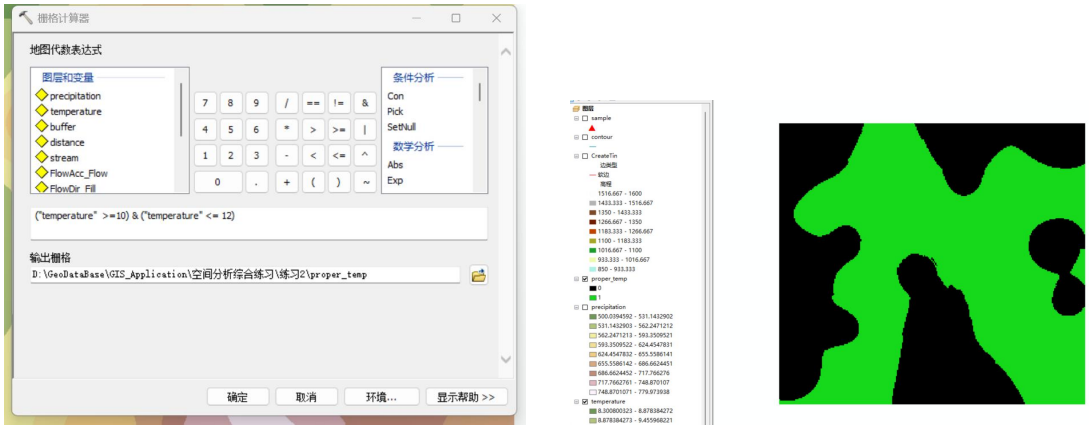
## (7) 导入观测点数据

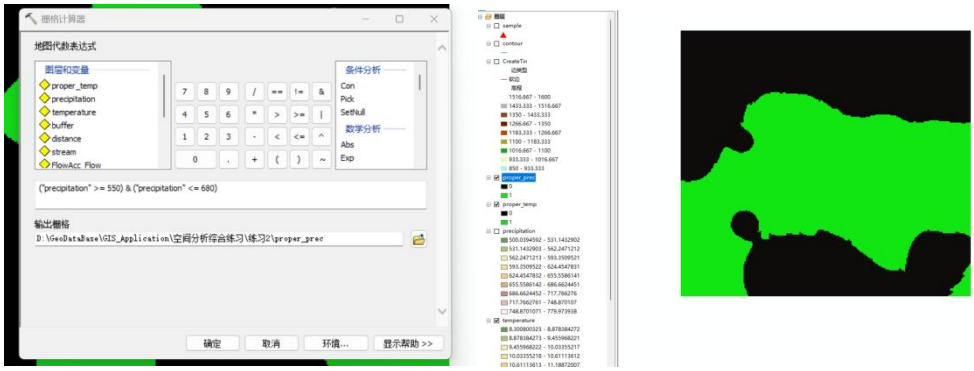



## (8) 分别内插生成“年平均温度”和“年降水量”表面数据

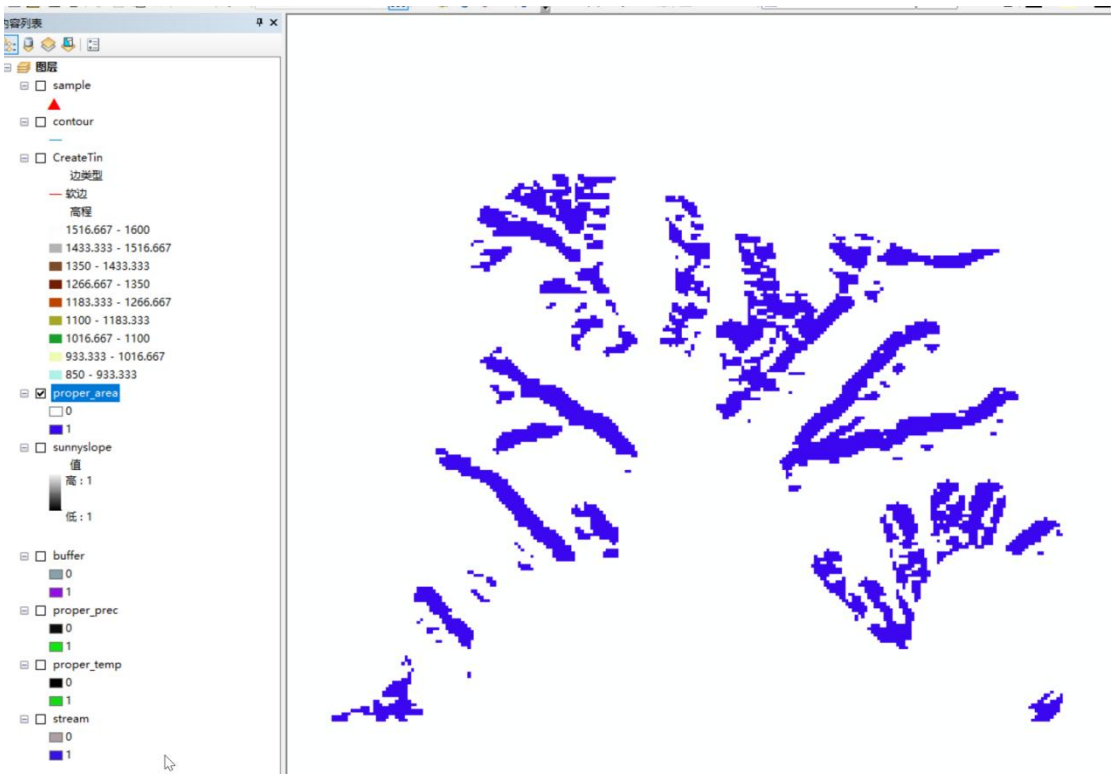


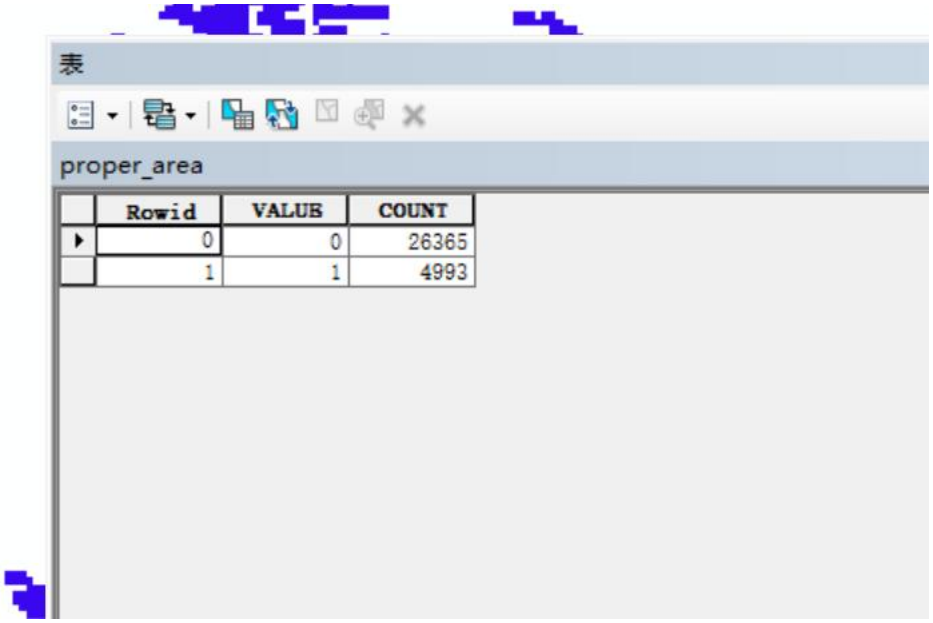
⑼ 根据药材生长条件，提取满足条件的温度和降水区域。




⑽ 提取满足 4 个条件的适合药材生长区域

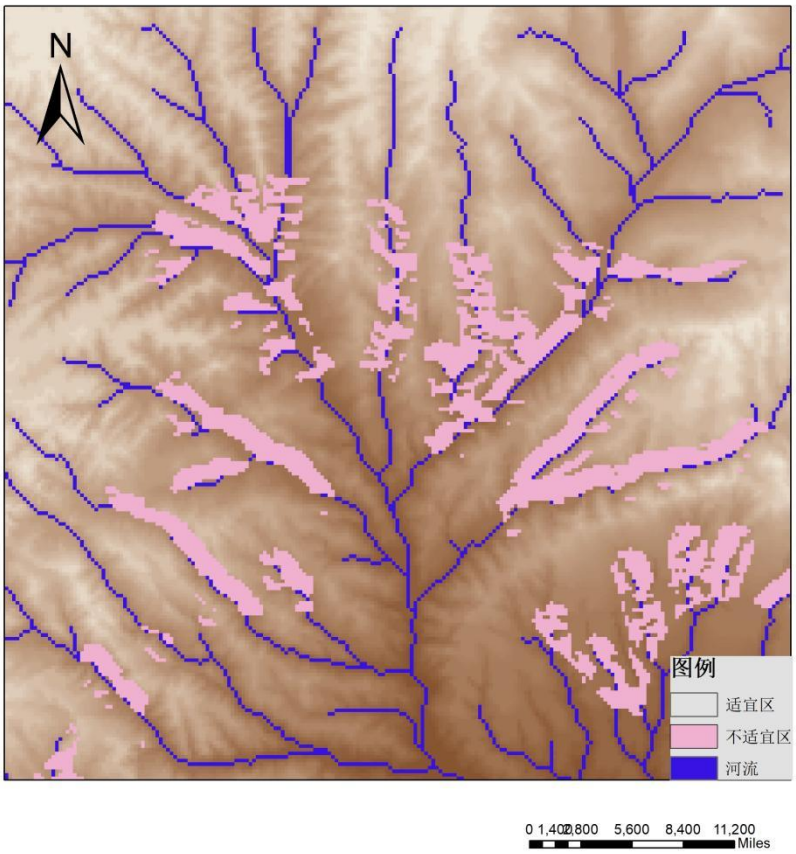




表

proper_area

| Rowid | VALUE | COUNT |
|---|---|---|
| 0 | 0 | 26365 |
| 1 | 1 | 4993 |

⑾ 制作专题地图

某种药材适合生长区域分布